

Elasticidad Dinamica - Departamento Papel

Modelo de betas por ventanas moviles

Filas SKU-FECHA: 4,396

SKUs Papel: 202

Fechas: 832

Filas con promocion: 266

Ventanas usadas:

Mensual: 1, 2, 3, 4

Trimestral: 1-3, 2-4, 3-5, 4-6

Semestral: 1-6, 2-7, 3-8, 4-9

Regresion estimada:

$\ln(\text{qty}) \sim \ln(\text{price})$

Insights principales para Papel

1. En Papel hay 202 SKUs y 4,396 observaciones SKU-FECHA; 6.05% de las filas tienen promocion detectada.
2. La ventana semestral es la mas estable: tiene mayor porcentaje de betas validas (45.48%) que mensual y trimestral.
3. El principal motivo de exclusion es $n < 3$, por lo que muchas ventanas no tienen suficientes observaciones para una regresion confiable.
4. Entre las betas validas, 52.84% son negativas; esas son las que siguen la relacion esperada de demanda precio-cantidad.
5. Las betas mensuales son utiles para detectar cambios bruscos, pero deben interpretarse con mas cuidado por menor numero de observaciones.

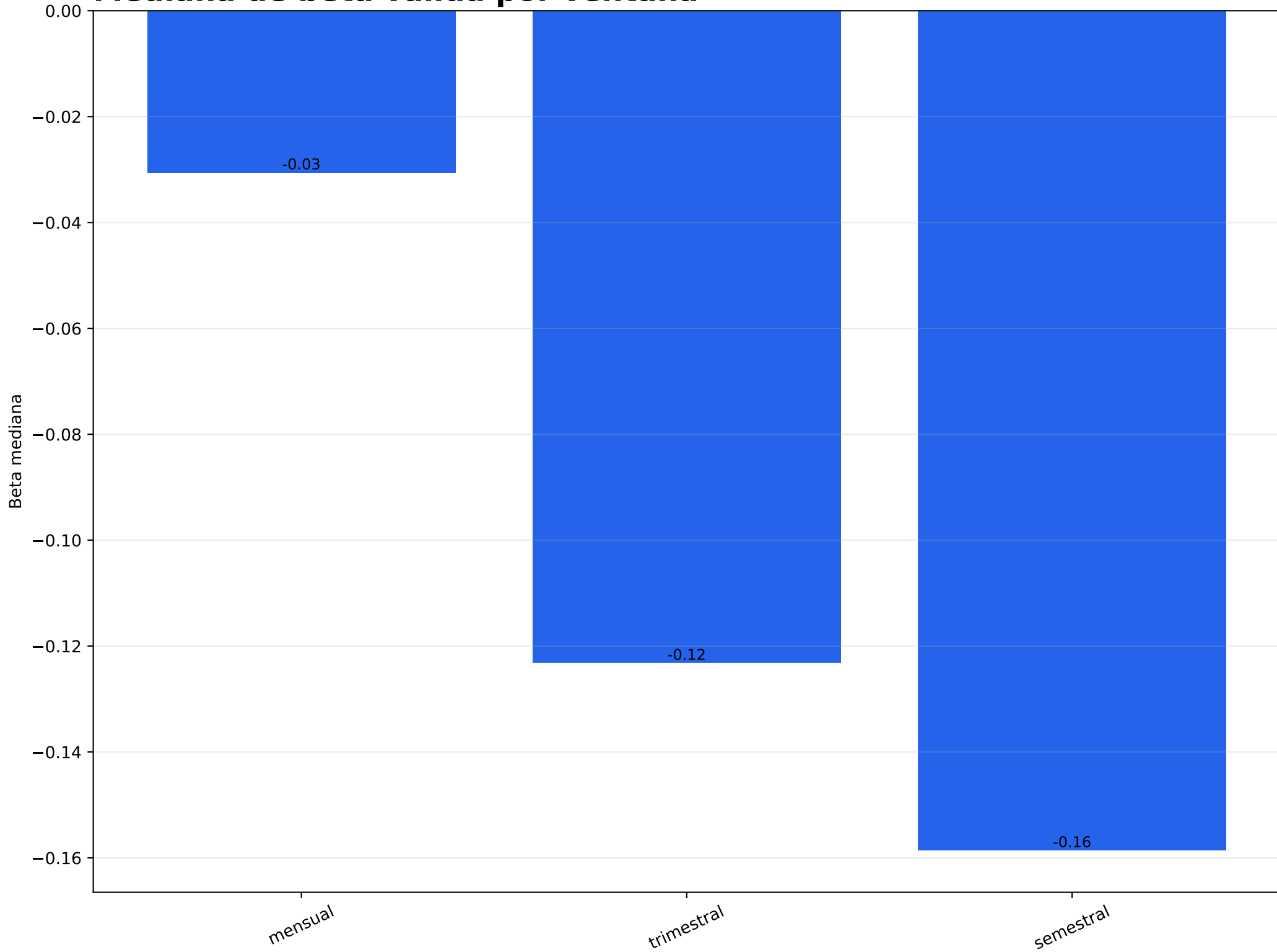
Betas validas por tipo de ventana

ventana	betas_validas	% valido	detalle
mensual	223	12.01%	223 validas de 1857 ventanas
semestral	951	45.48%	951 validas de 2091 ventanas
trimestral	660	29.64%	660 validas de 2227 ventanas

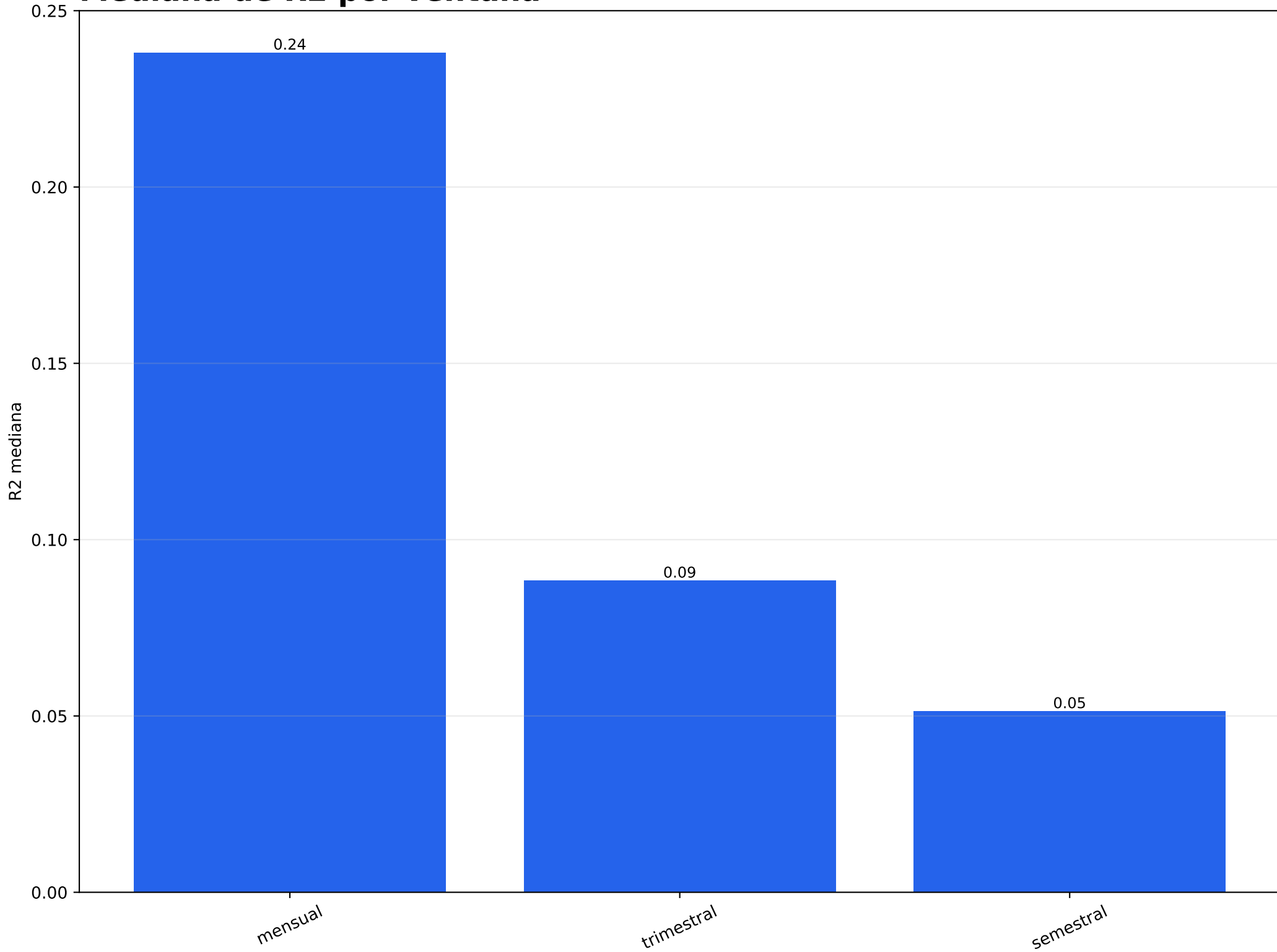
Razones de exclusion

razon	ventanas	%
n<3	2710	62.43%
precio constante	871	20.06%
datos insuficientes	615	14.17%
qty constante	145	3.34%

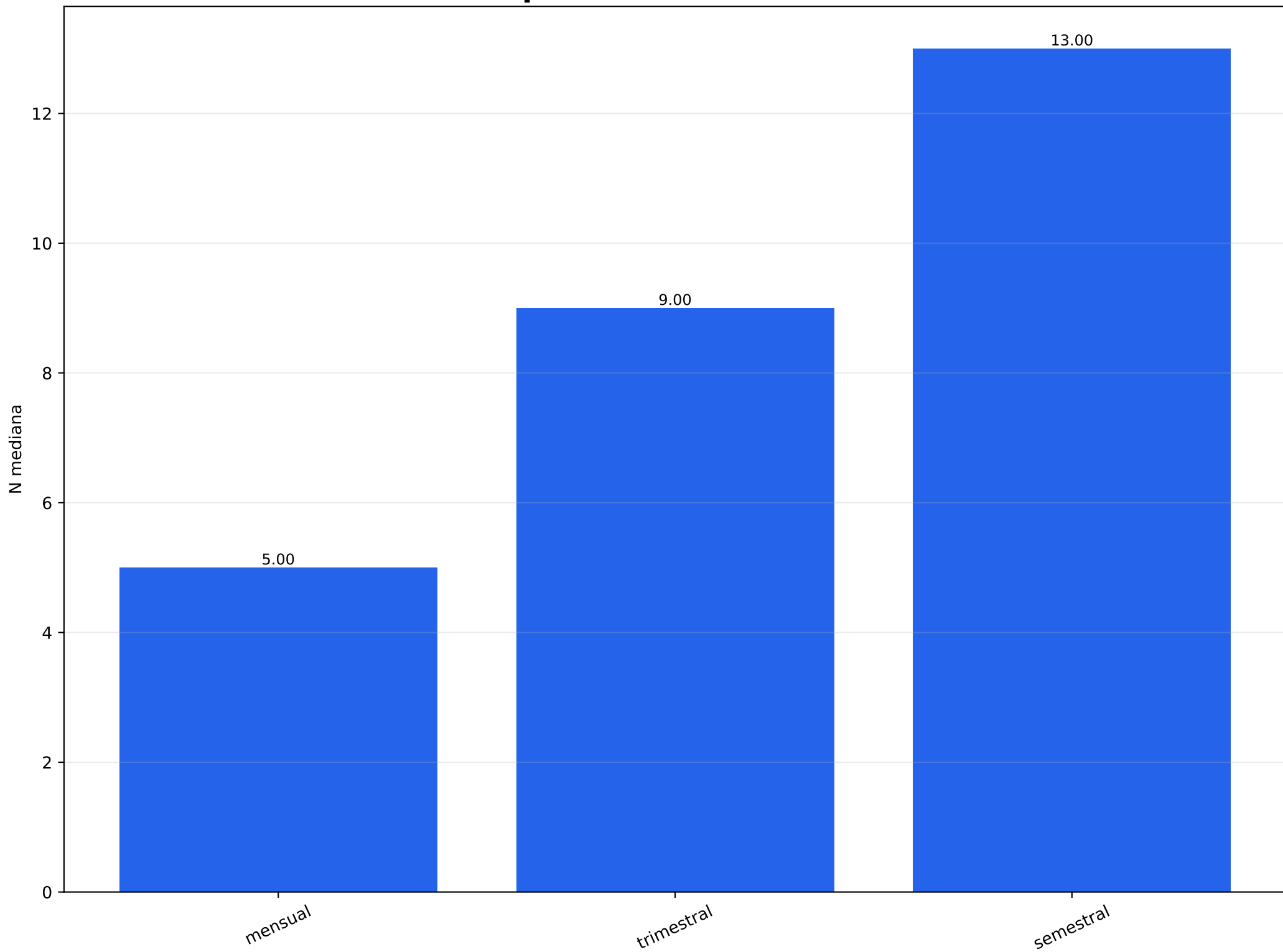
Mediana de beta valida por ventana



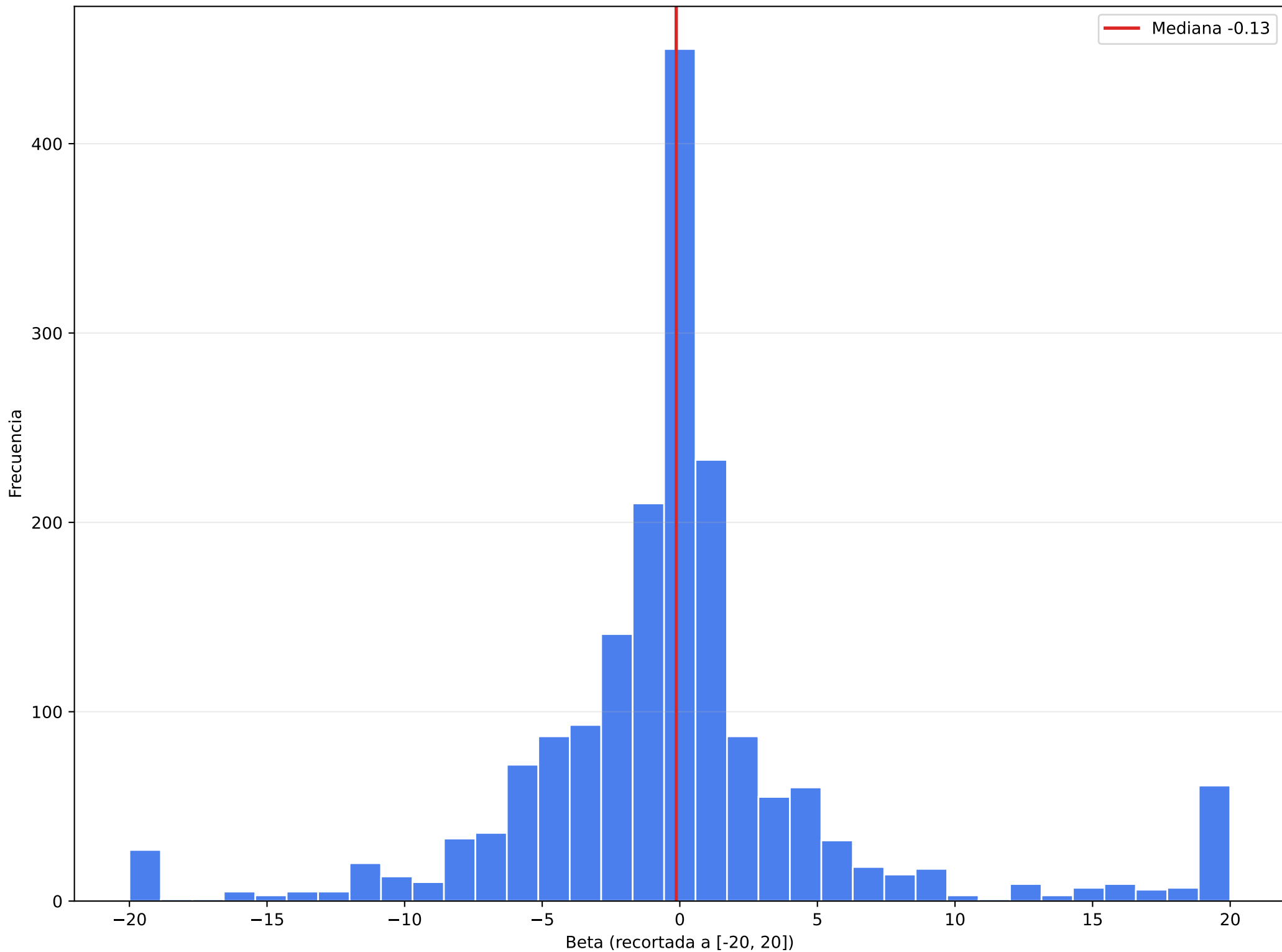
Mediana de R2 por ventana



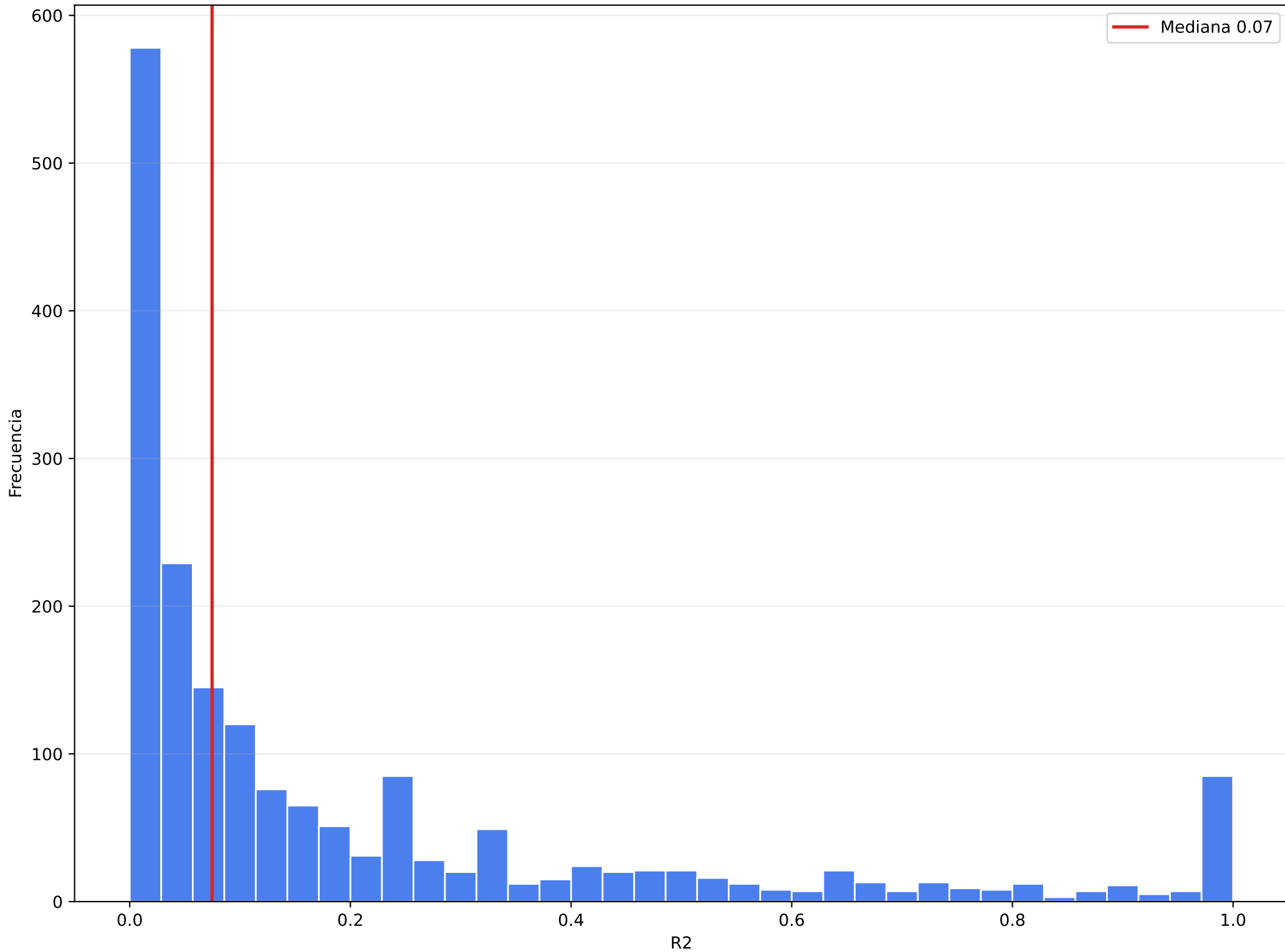
Mediana de observaciones por ventana

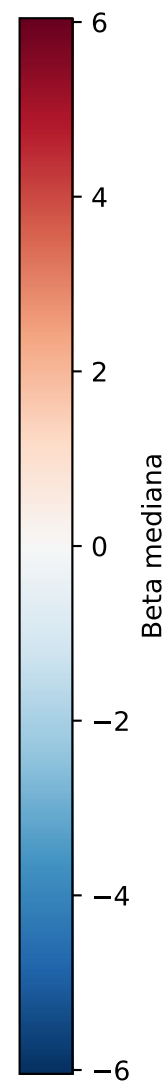


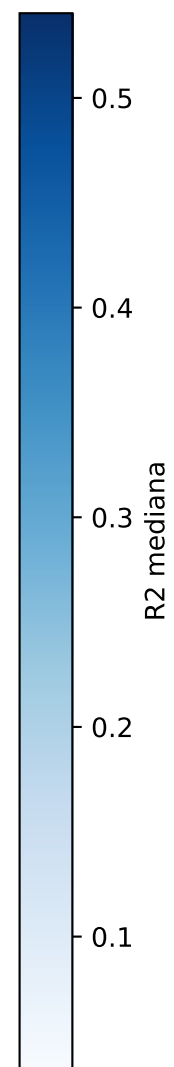
Distribucion de betas validas



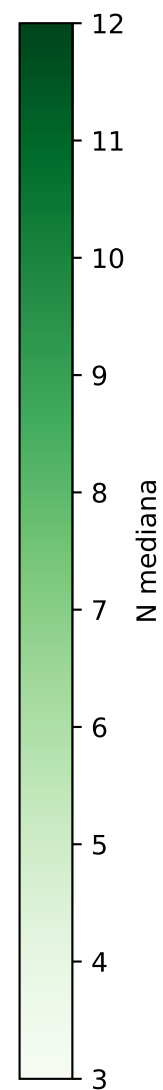
Distribucion de R2 en betas validas

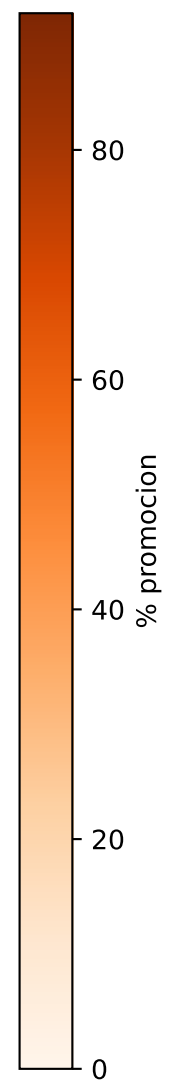




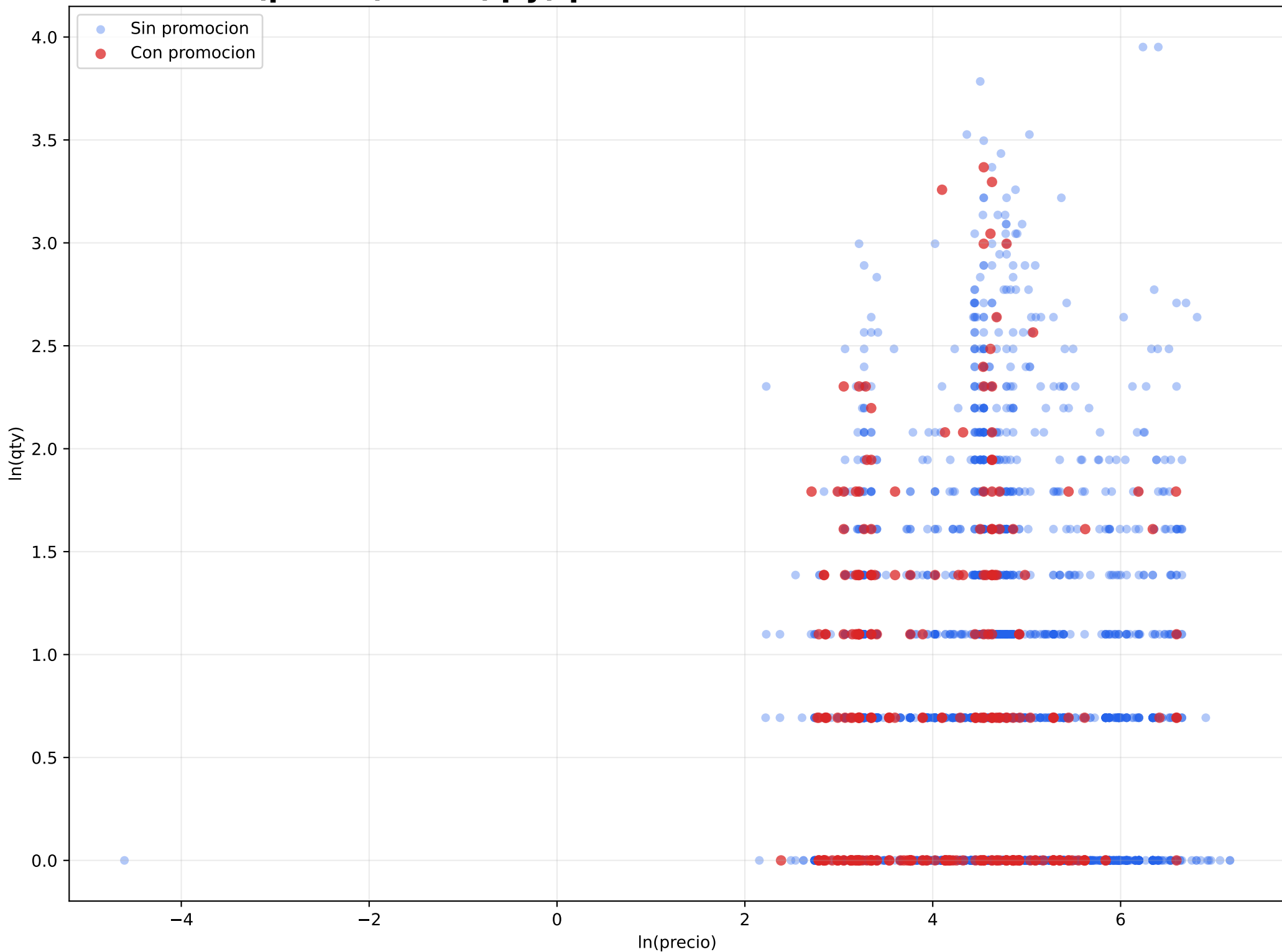
[illegible]

The heatmap consists of 3 rows and 12 columns. The color scale ranges from dark green (high values) to light green/white (low values). The bottom row (row 3) is predominantly white, indicating low values across most columns. The top two rows (rows 1 and 2) show higher values, represented by various shades of green. The highest values (darkest green) are concentrated in the middle columns (columns 4, 5, and 6) of the top two rows.



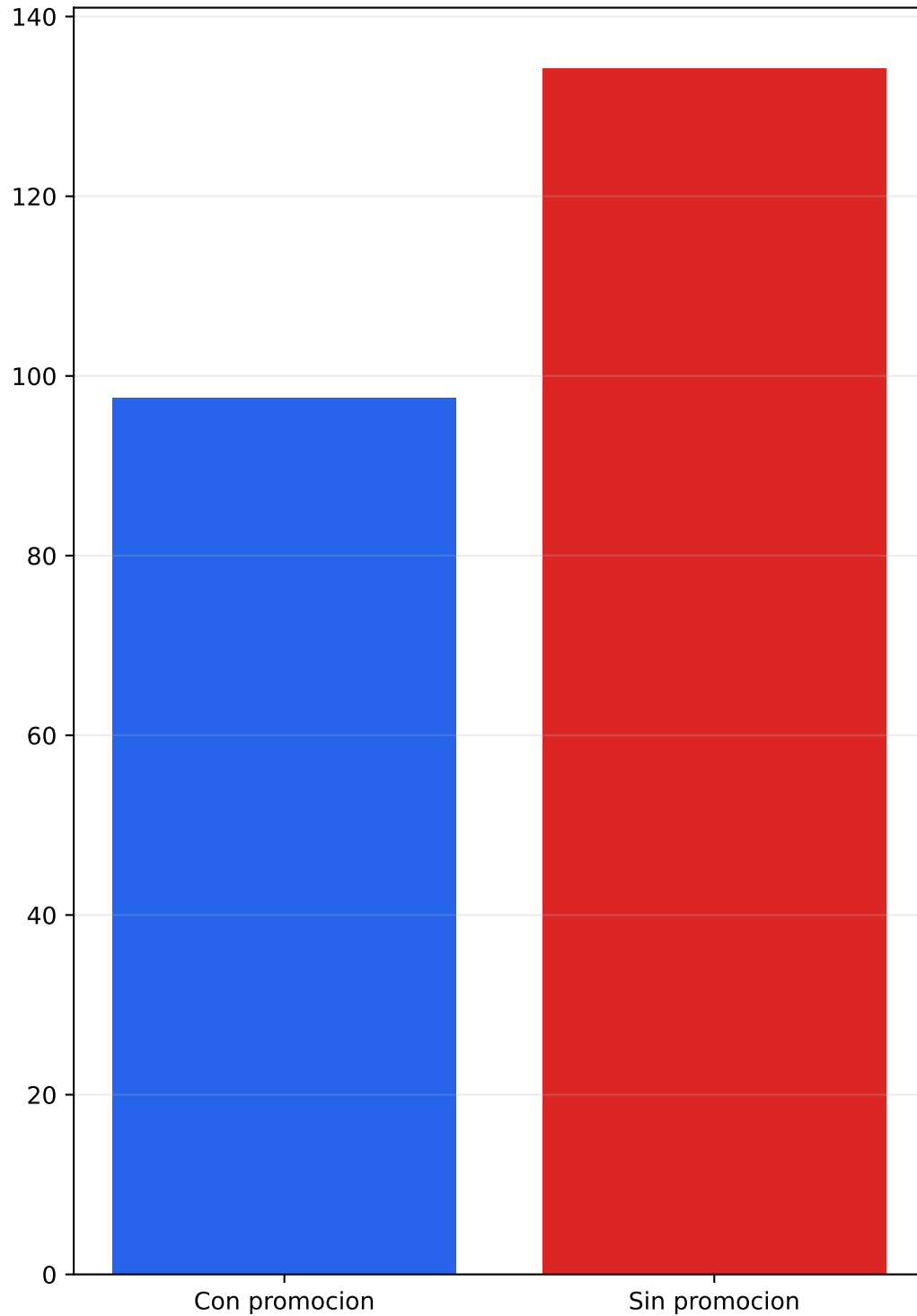


Relacion $\ln(\text{precio})$ vs $\ln(\text{qty})$ por observacion SKU-FECHA

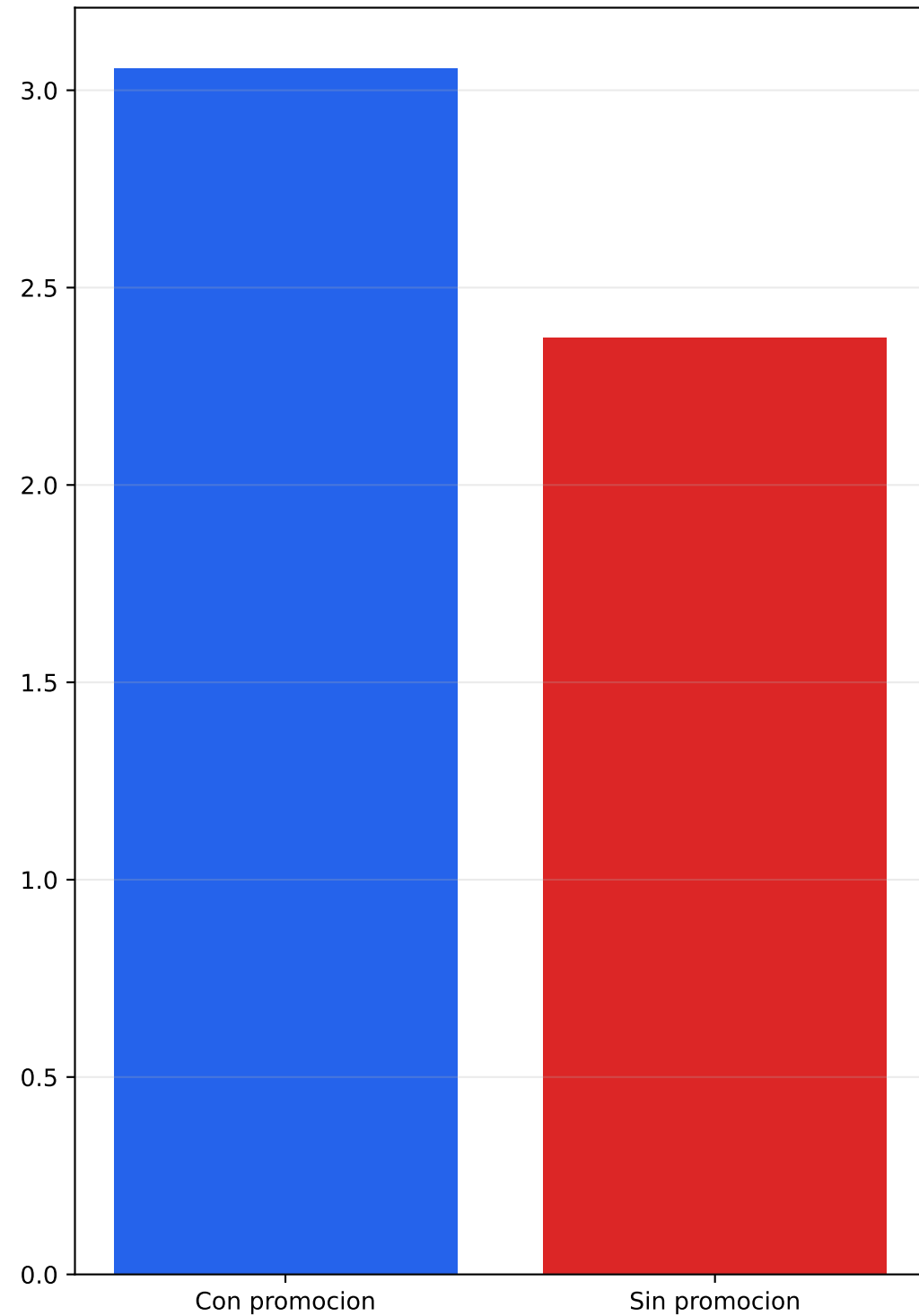


Comparativo con promocion vs sin promocion

Precio promedio



Qty promedio



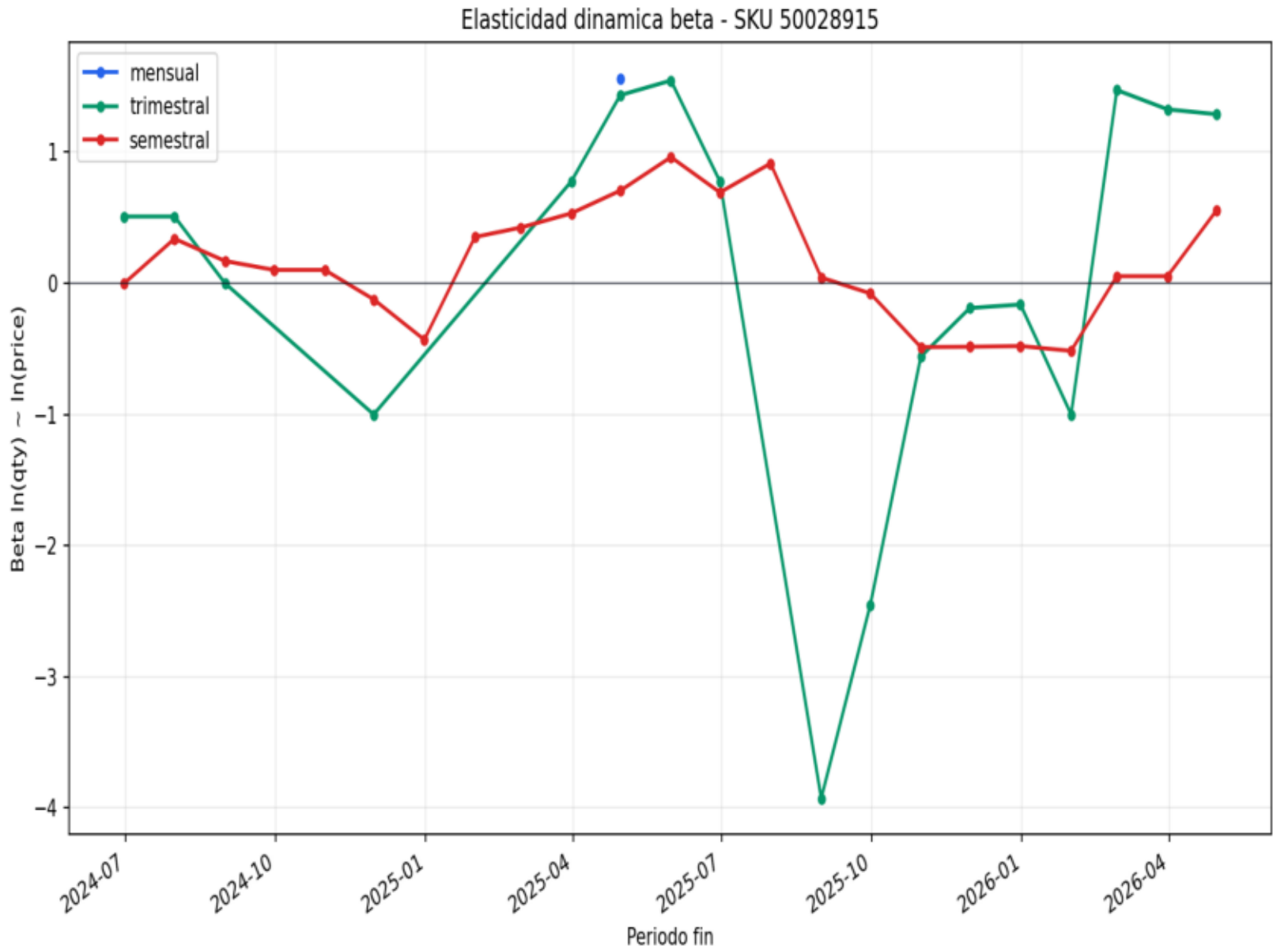
Top 20 SKUs de Papel con mas betas validas

SKU	betas_validas	% validas	detalle
50084493	68	88.31%	68 validas de 77 ventanas
50106204	64	83.12%	64 validas de 77 ventanas
50077229	62	80.52%	62 validas de 77 ventanas
50046029	61	79.22%	61 validas de 77 ventanas
50056239	52	70.27%	52 validas de 74 ventanas
50033183	49	73.13%	49 validas de 67 ventanas
50062268	49	63.64%	49 validas de 77 ventanas
50096451	47	61.84%	47 validas de 76 ventanas
50065845	45	60.00%	45 validas de 75 ventanas
50062267	44	66.67%	44 validas de 66 ventanas
50073101	43	58.11%	43 validas de 74 ventanas
50062025	42	57.53%	42 validas de 73 ventanas
50066990	42	56.00%	42 validas de 75 ventanas
50046028	42	54.55%	42 validas de 77 ventanas
50028915	41	56.94%	41 validas de 72 ventanas
50086321	41	54.67%	41 validas de 75 ventanas
50062269	40	52.63%	40 validas de 76 ventanas
50050823	39	52.70%	39 validas de 74 ventanas
50063368	39	51.32%	39 validas de 76 ventanas
50046030	37	48.05%	37 validas de 77 ventanas

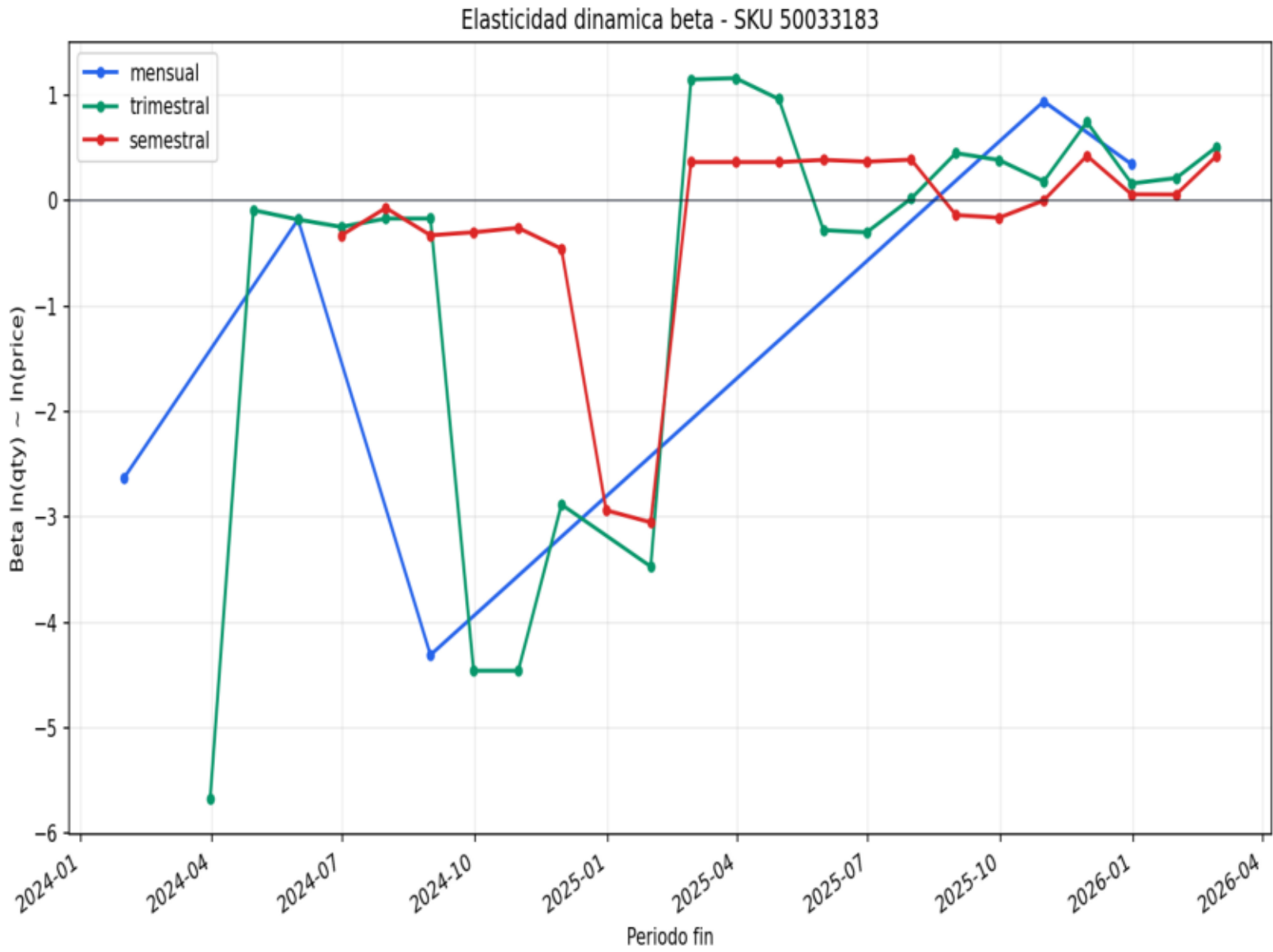
Cambios importantes de beta por SKU y ventana

SKU	tipo_ventana	cambios	cambios_signo	max_abs_delta	mediana_abs_delta
50046029	trimestral	19	6	45.25	3.14
50077229	mensual	15	6	134.68	10.40
50077229	trimestral	14	2	9.76	2.33
50096451	trimestral	14	5	8.21	3.77
50106204	mensual	13	10	23.30	5.37
50084493	mensual	13	8	15.88	2.22
50050823	trimestral	13	6	13.91	2.21
50106204	trimestral	13	10	3.81	1.42
50046029	mensual	12	5	222.93	7.49
50056239	trimestral	12	7	15.19	3.63
50065845	semestral	12	2	8.21	2.74
50062268	trimestral	11	3	5534.82	3.79
50066990	trimestral	11	4	15.86	3.86
50050823	semestral	11	6	13.80	2.88
50046029	semestral	11	4	10.05	2.25
50073101	semestral	11	7	7.16	1.56
50065845	trimestral	11	2	7.12	2.33
50062267	trimestral	10	4	25.59	4.74
50084493	trimestral	10	8	12.22	0.94
50073101	trimestral	10	3	7.59	3.01

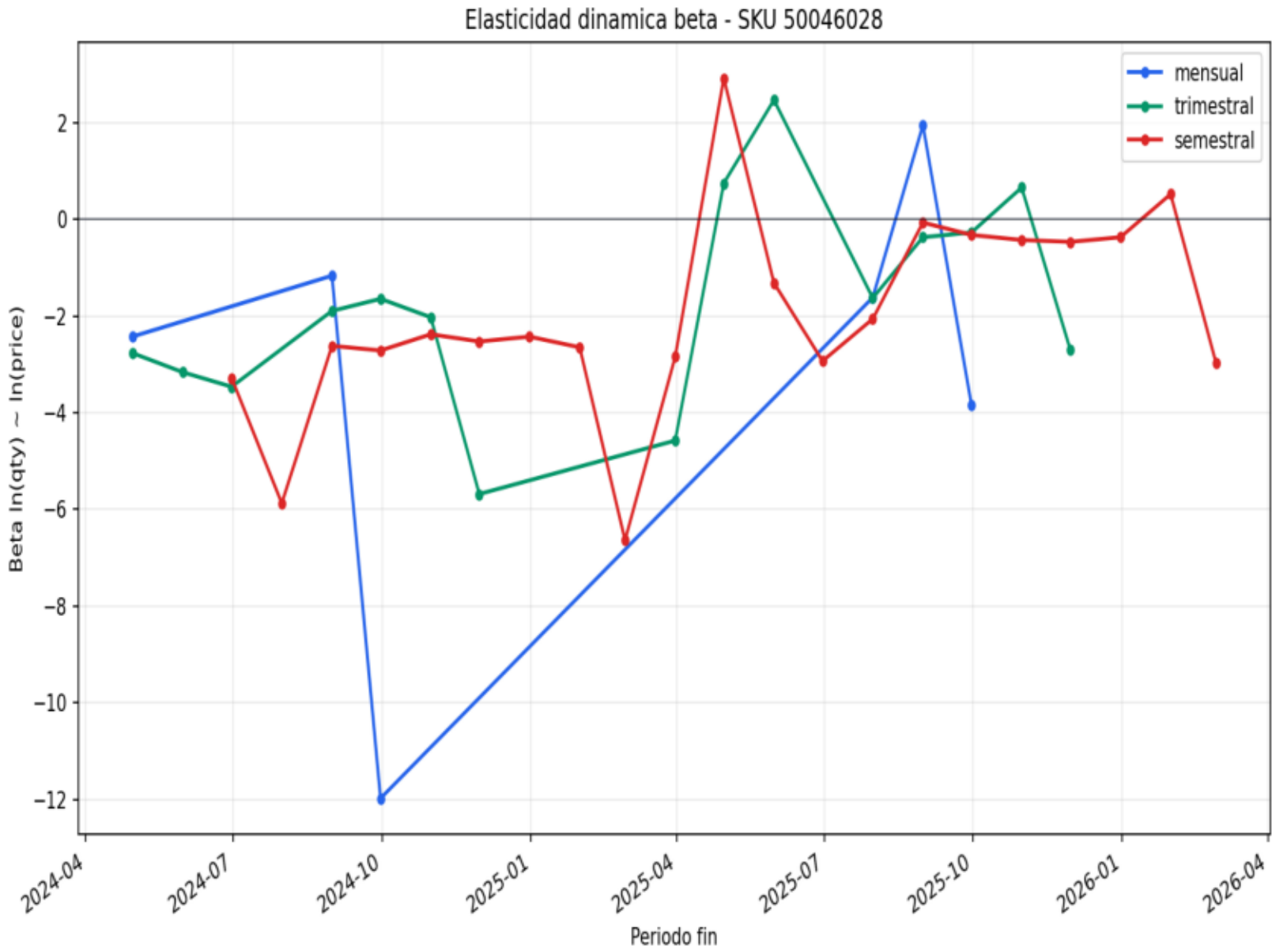
Beta mensual, trimestral y semestral - SKU 50028915



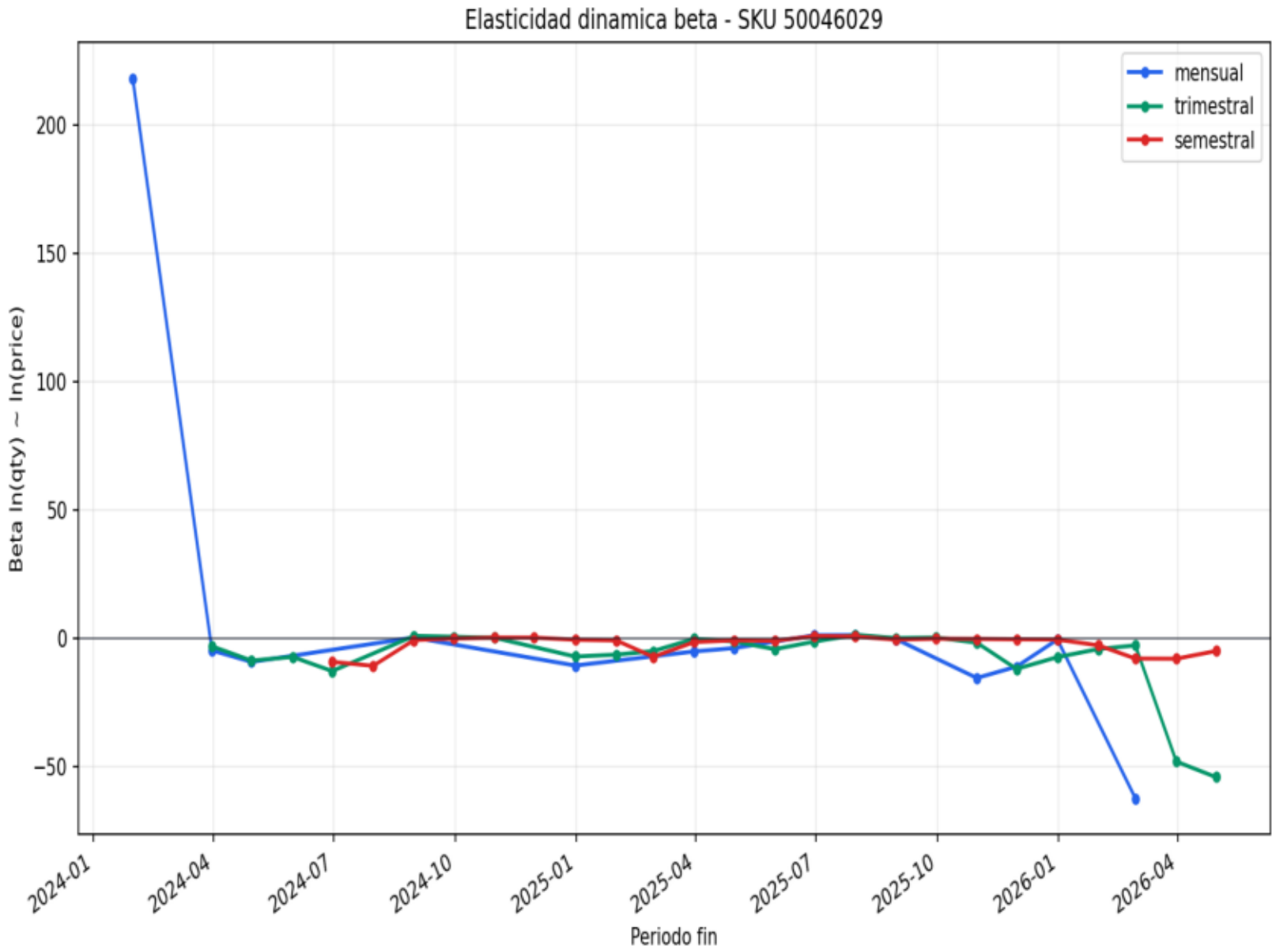
Beta mensual, trimestral y semestral - SKU 50033183



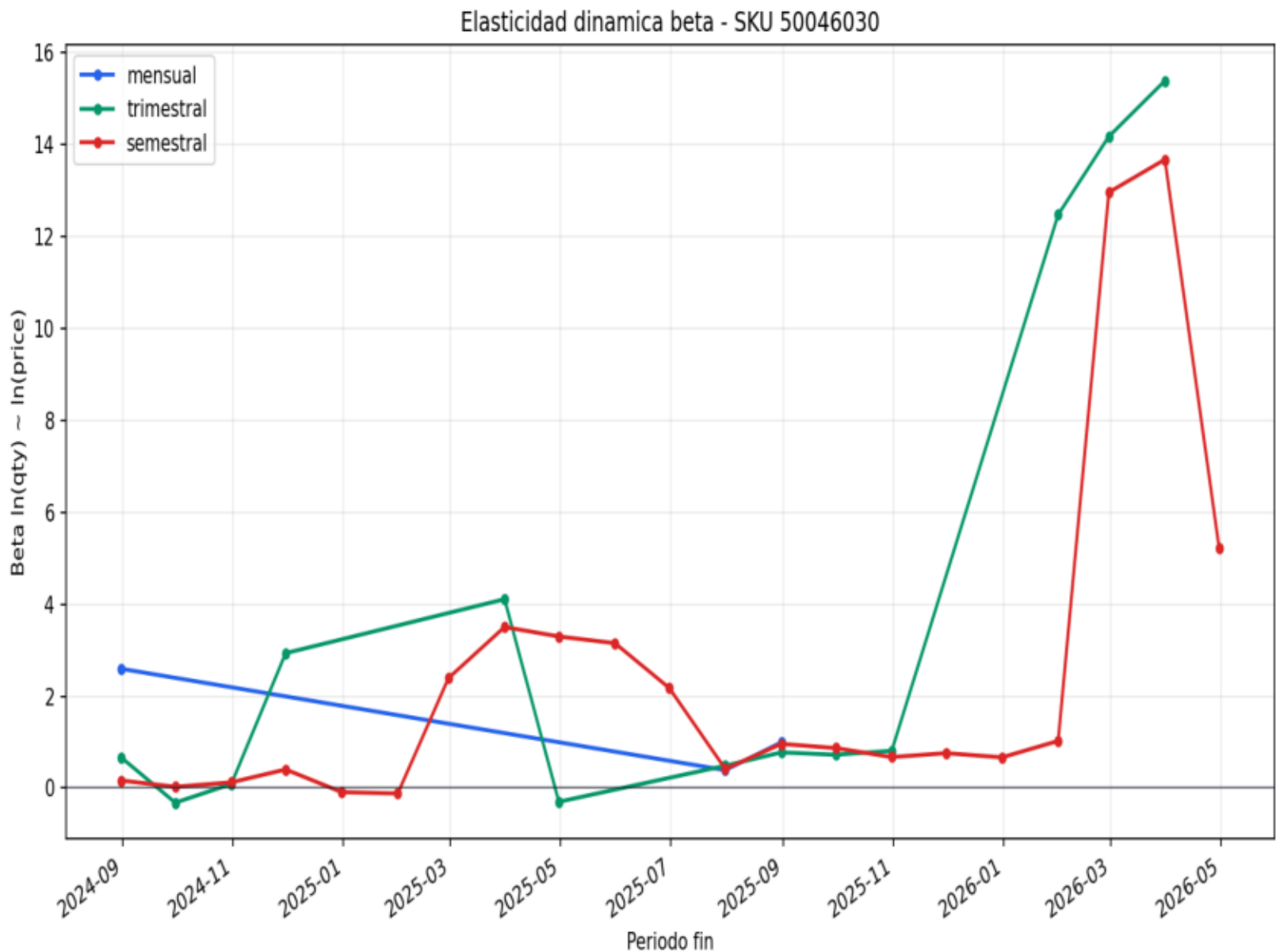
Beta mensual, trimestral y semestral - SKU 50046028



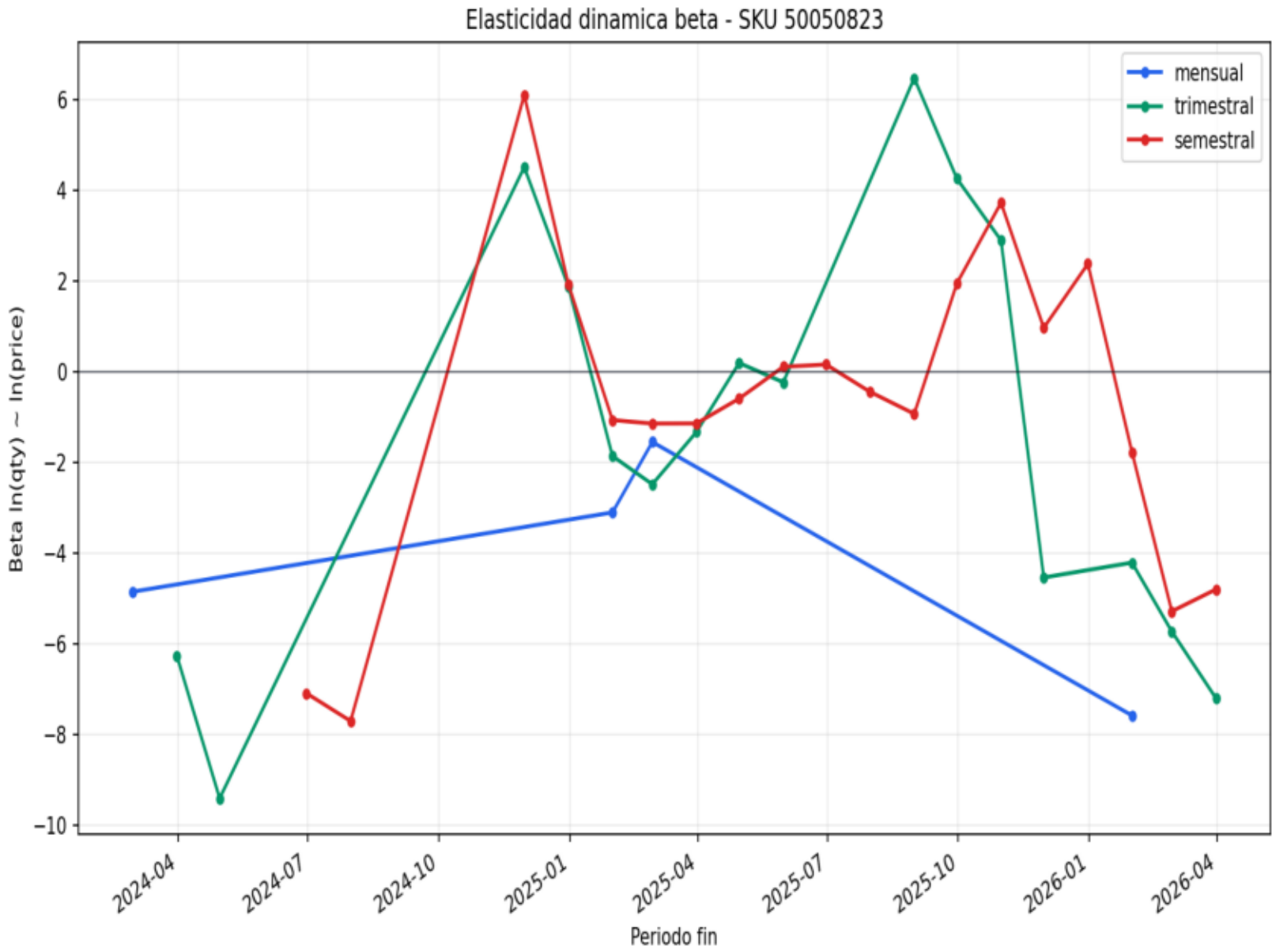
Beta mensual, trimestral y semestral - SKU 50046029



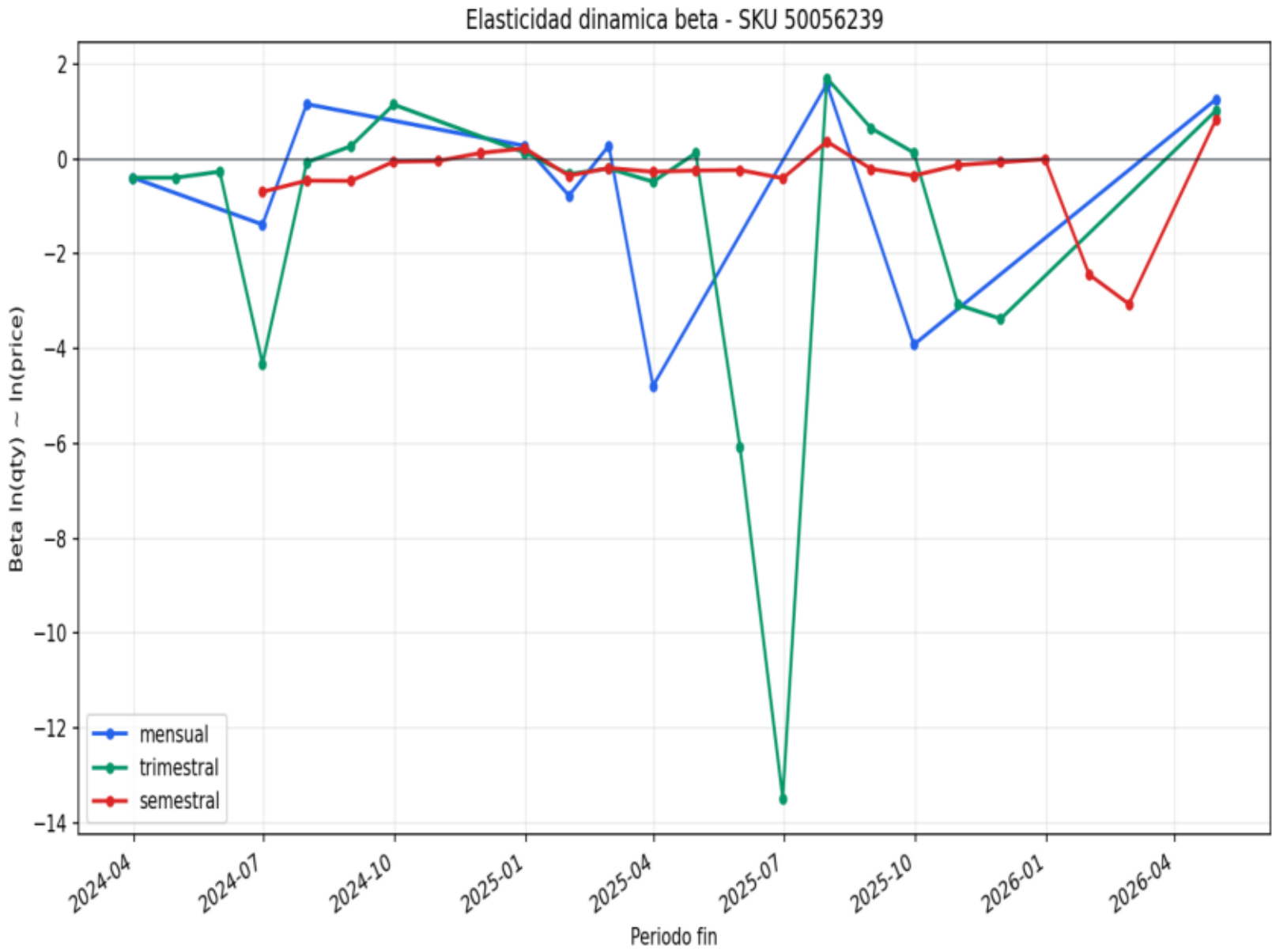
Beta mensual, trimestral y semestral - SKU 50046030



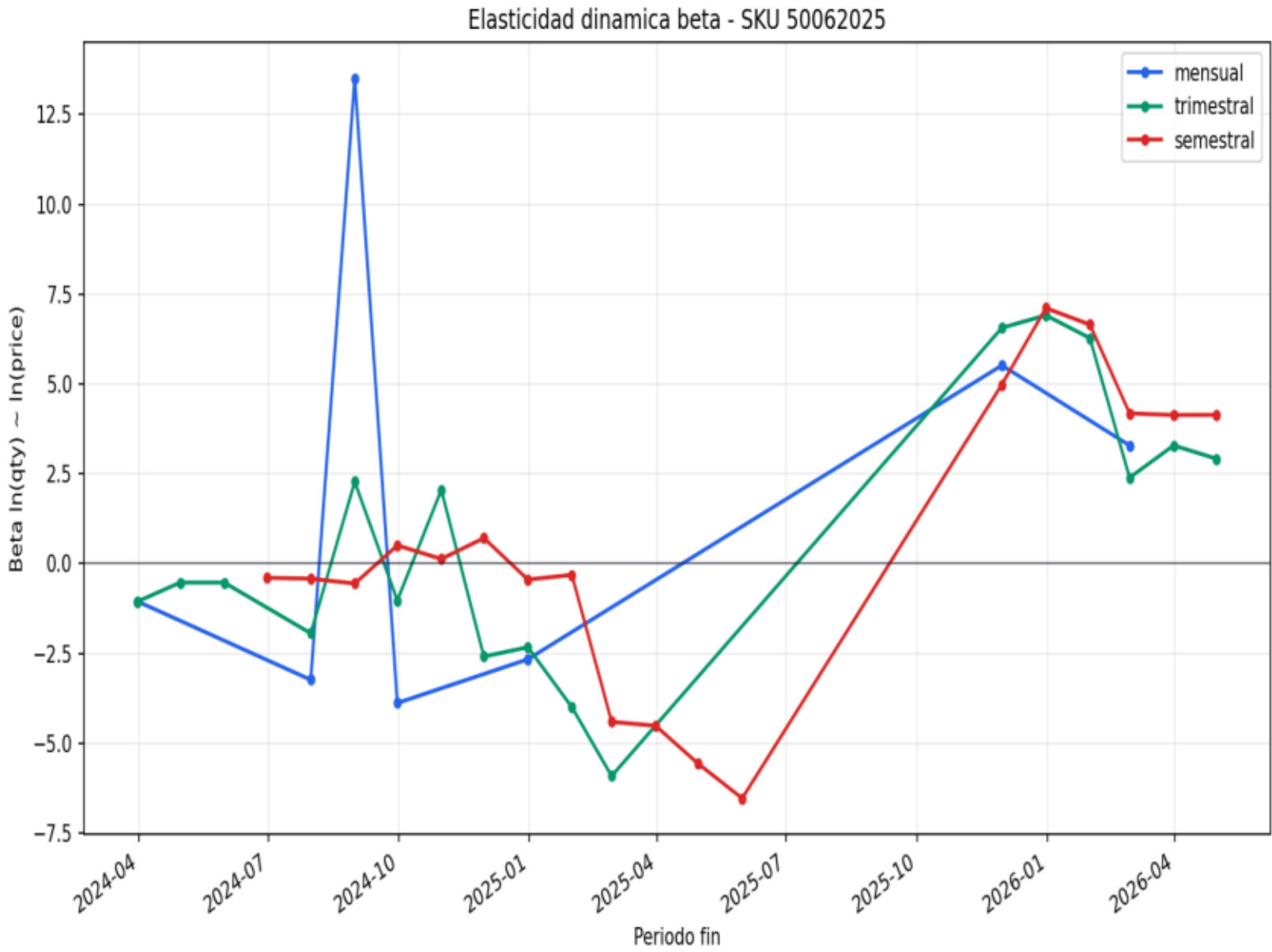
Beta mensual, trimestral y semestral - SKU 50050823



Beta mensual, trimestral y semestral - SKU 50056239

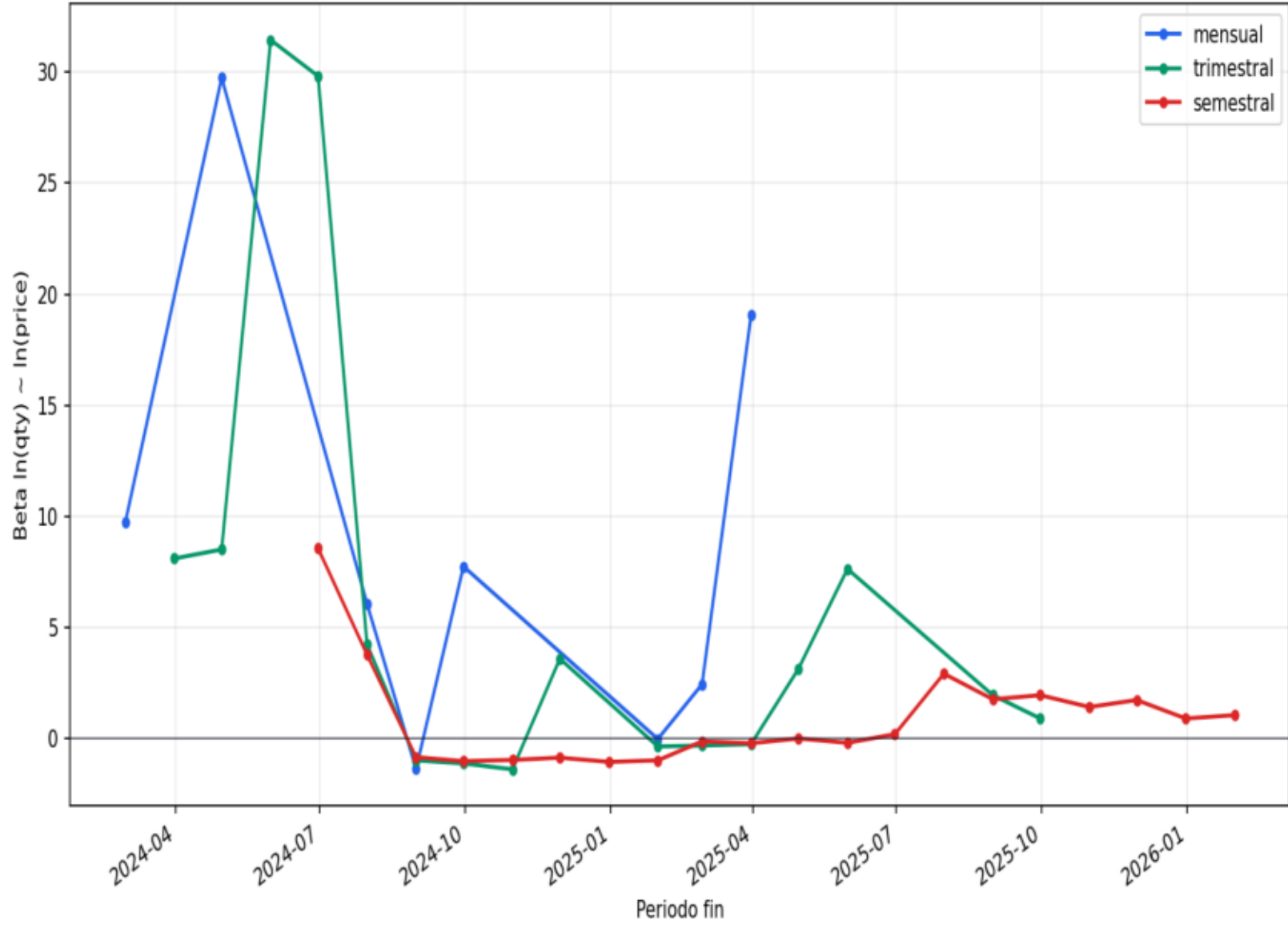


Beta mensual, trimestral y semestral - SKU 50062025

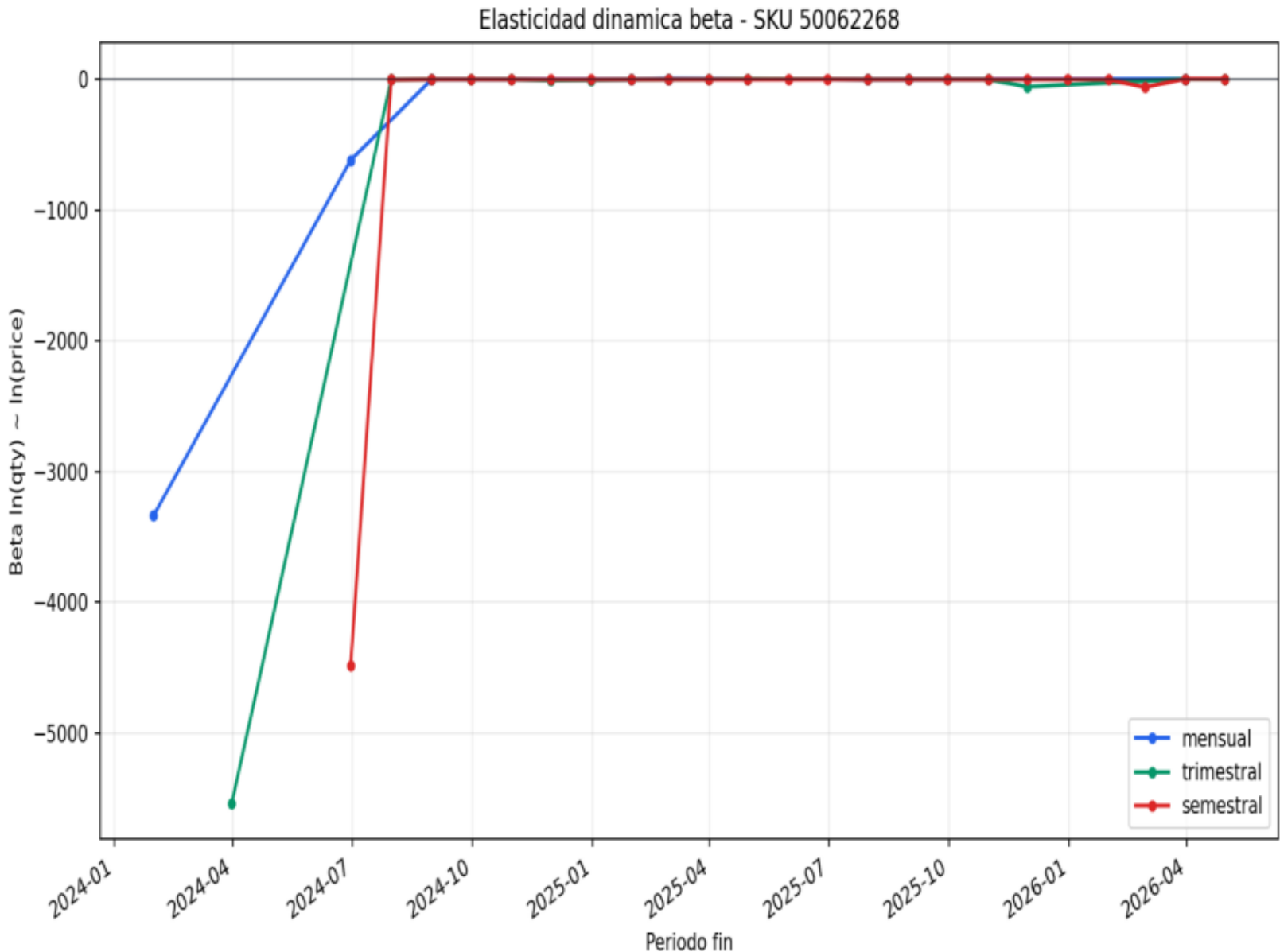


Beta mensual, trimestral y semestral - SKU 50062267

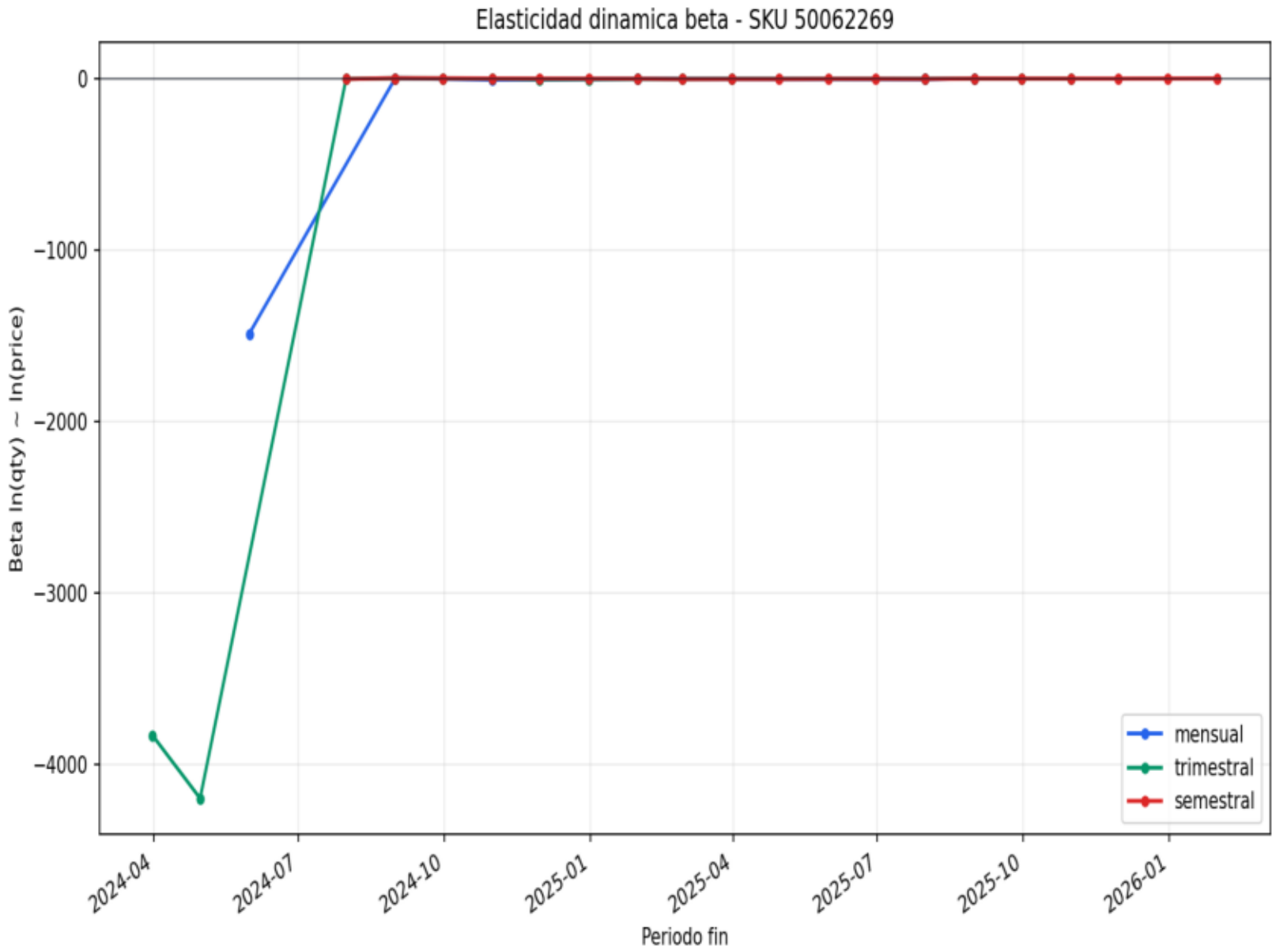
Elasticidad dinamica beta - SKU 50062267



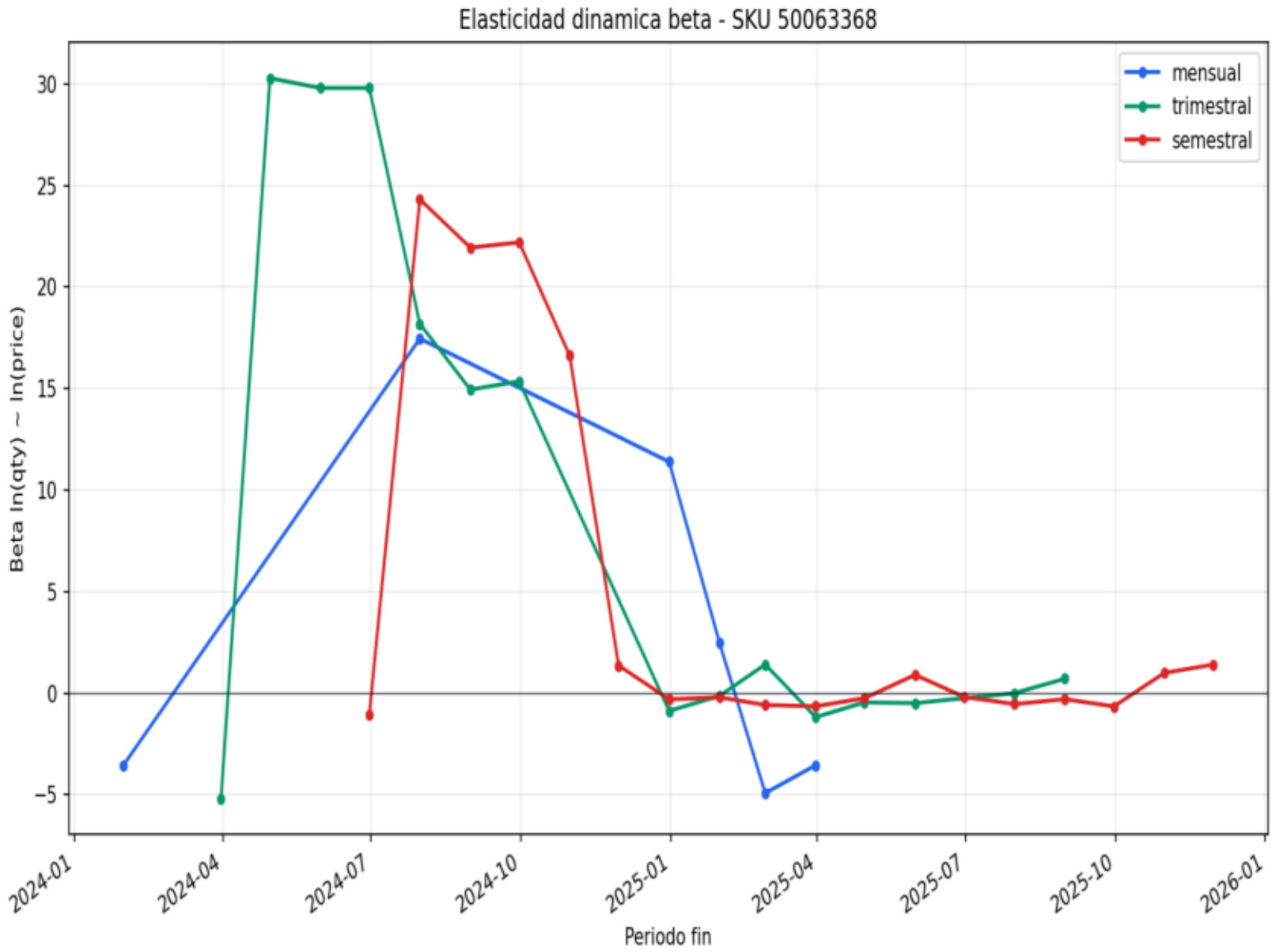
Beta mensual, trimestral y semestral - SKU 50062268



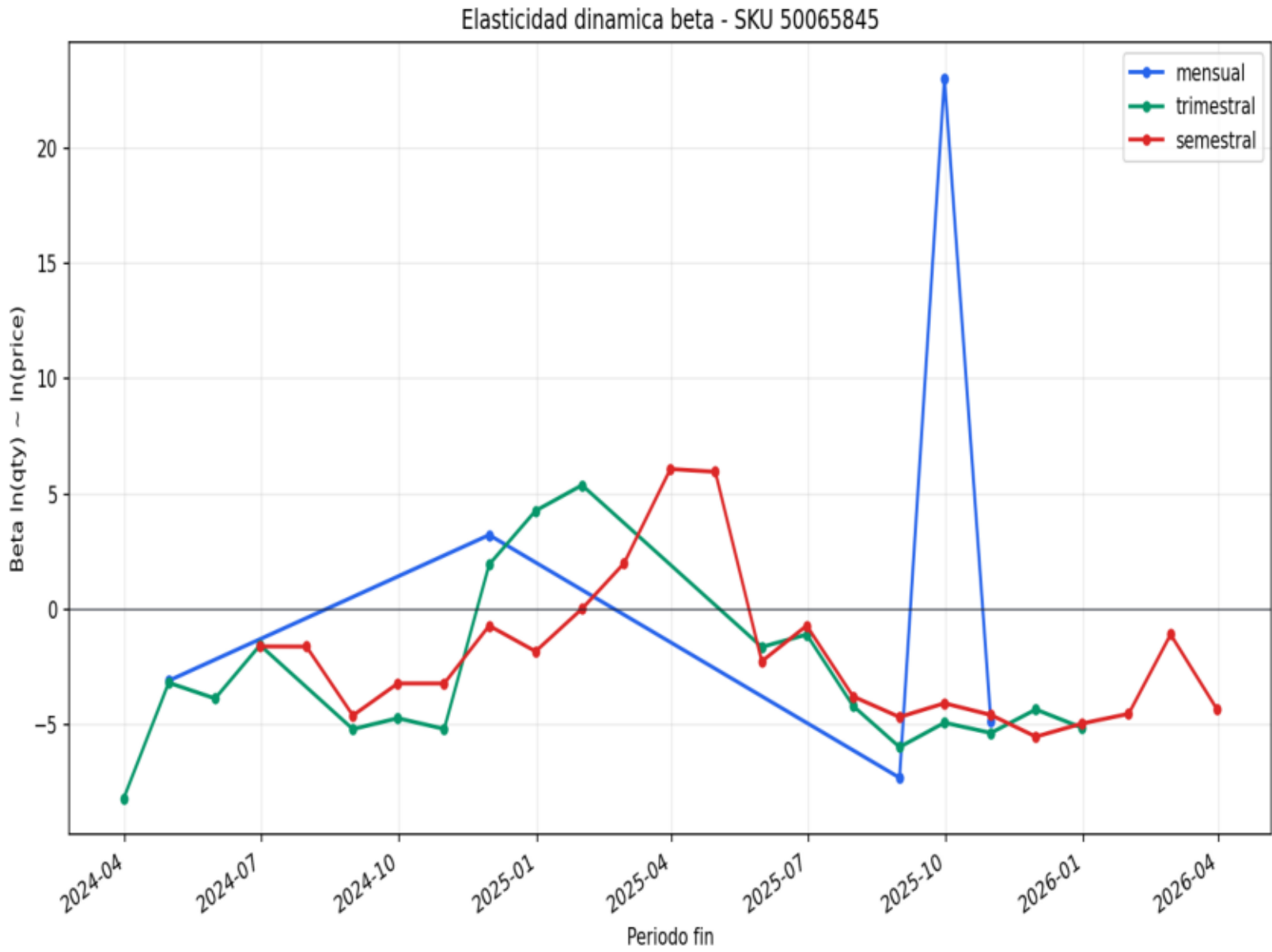
Beta mensual, trimestral y semestral - SKU 50062269



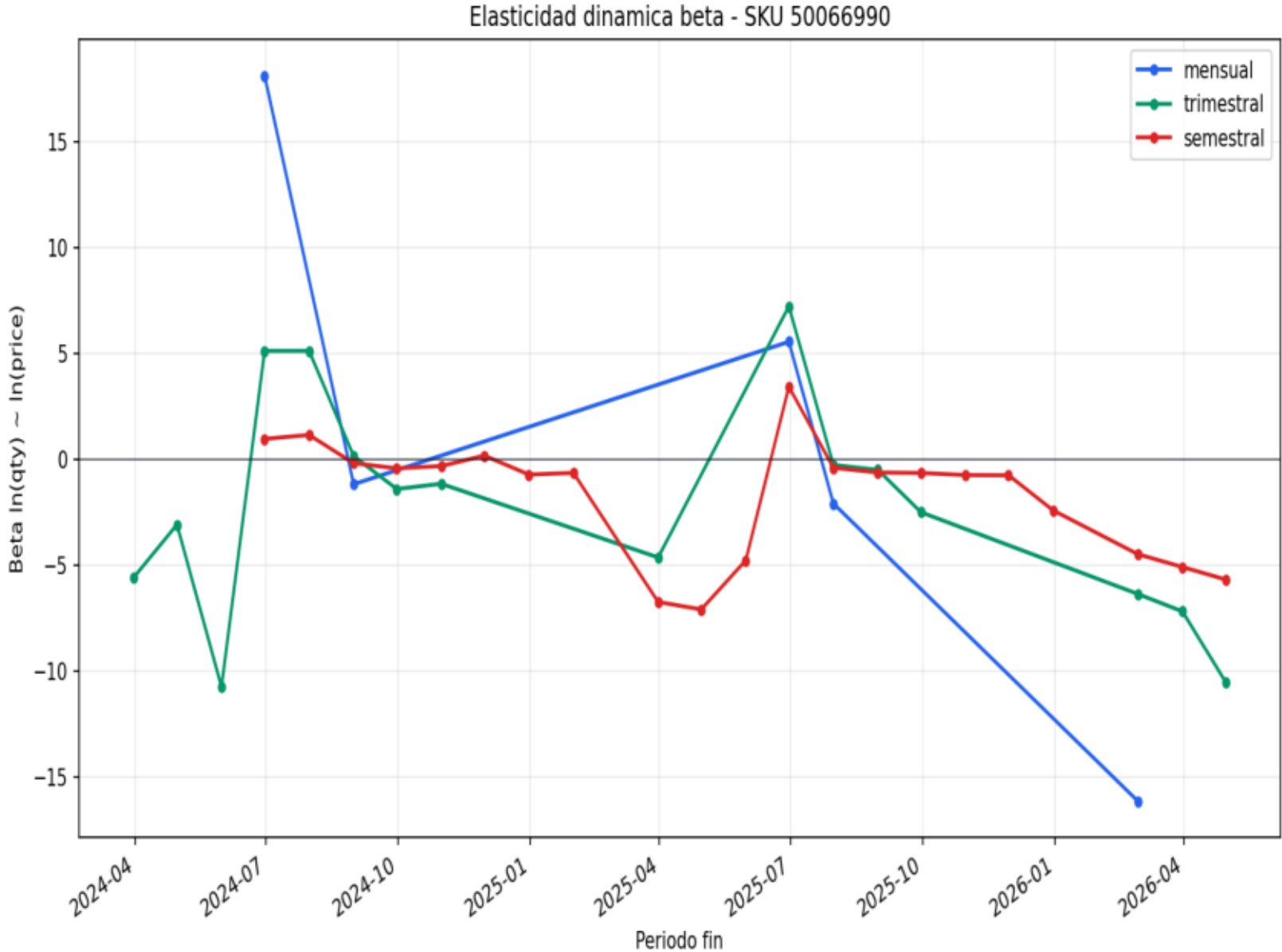
Beta mensual, trimestral y semestral - SKU 50063368



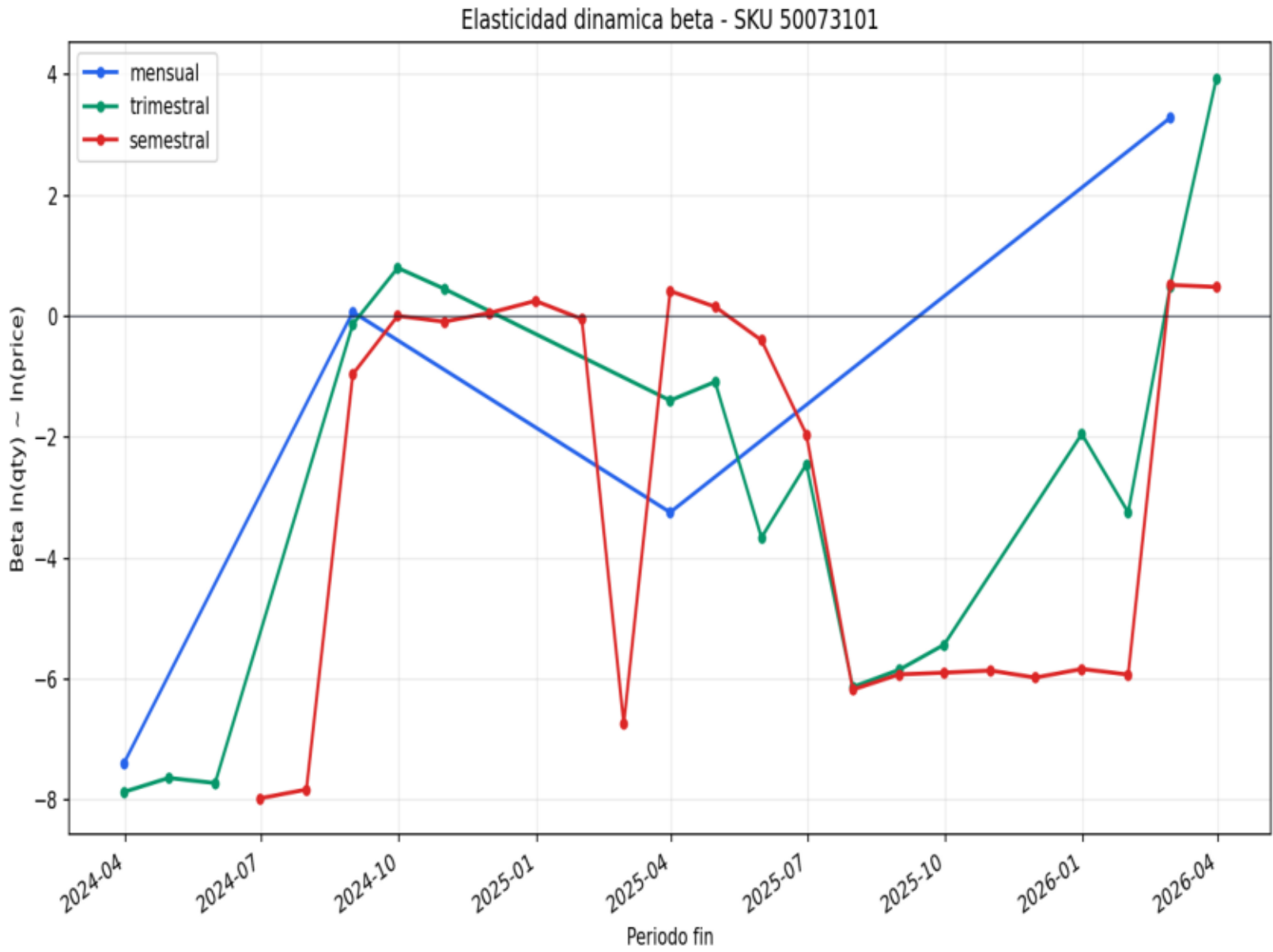
Beta mensual, trimestral y semestral - SKU 50065845



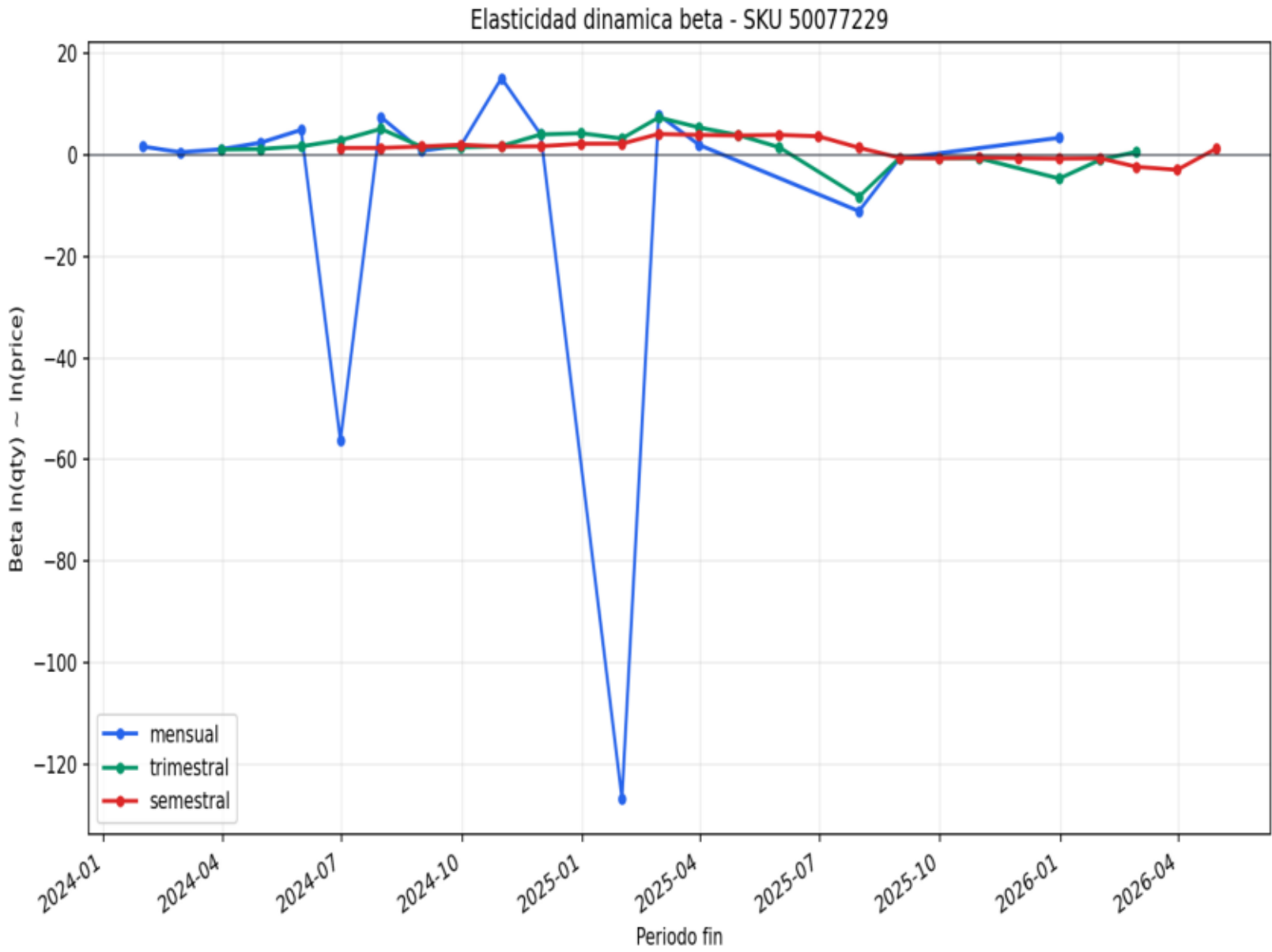
Beta mensual, trimestral y semestral - SKU 50066990



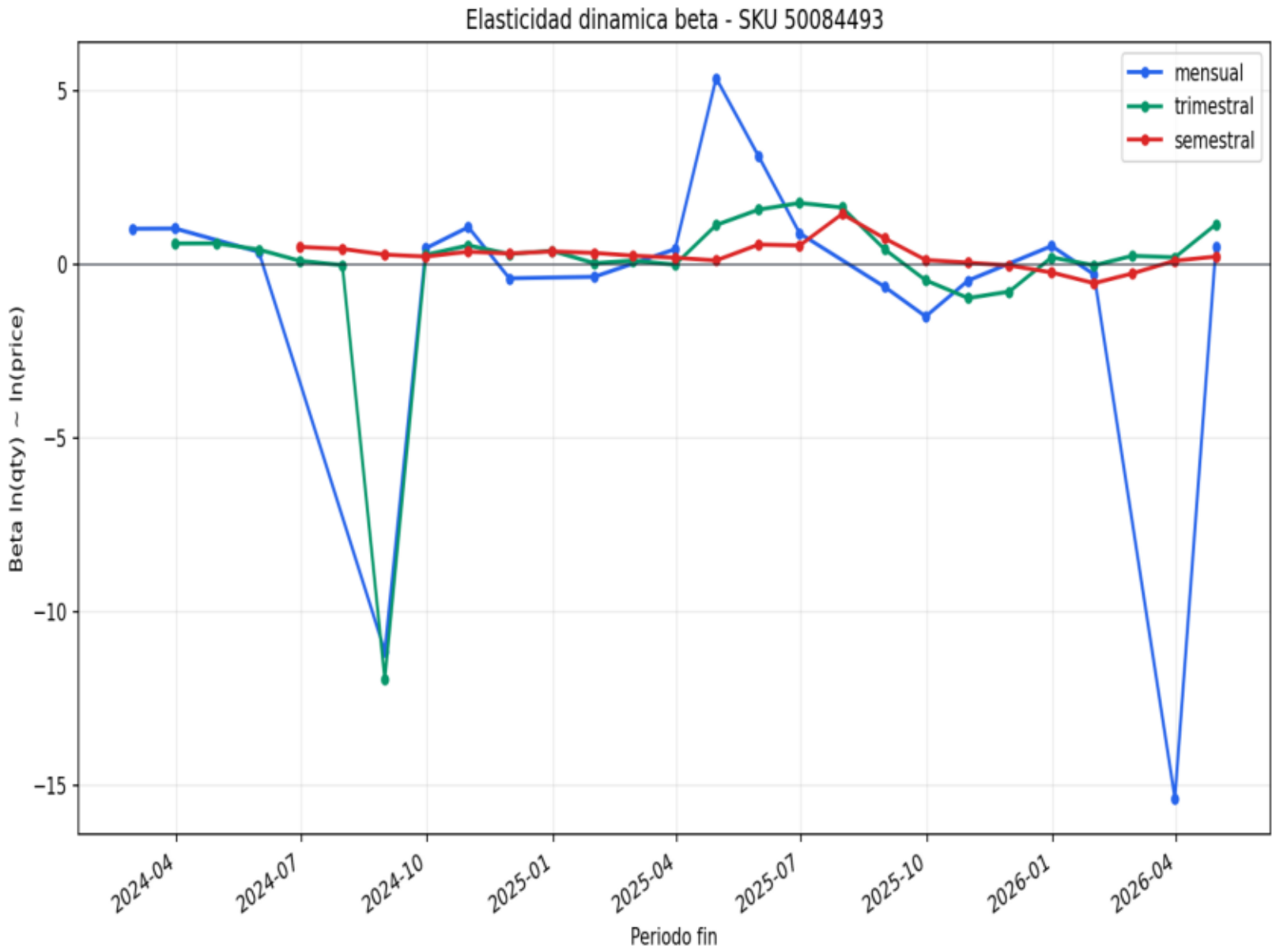
Beta mensual, trimestral y semestral - SKU 50073101



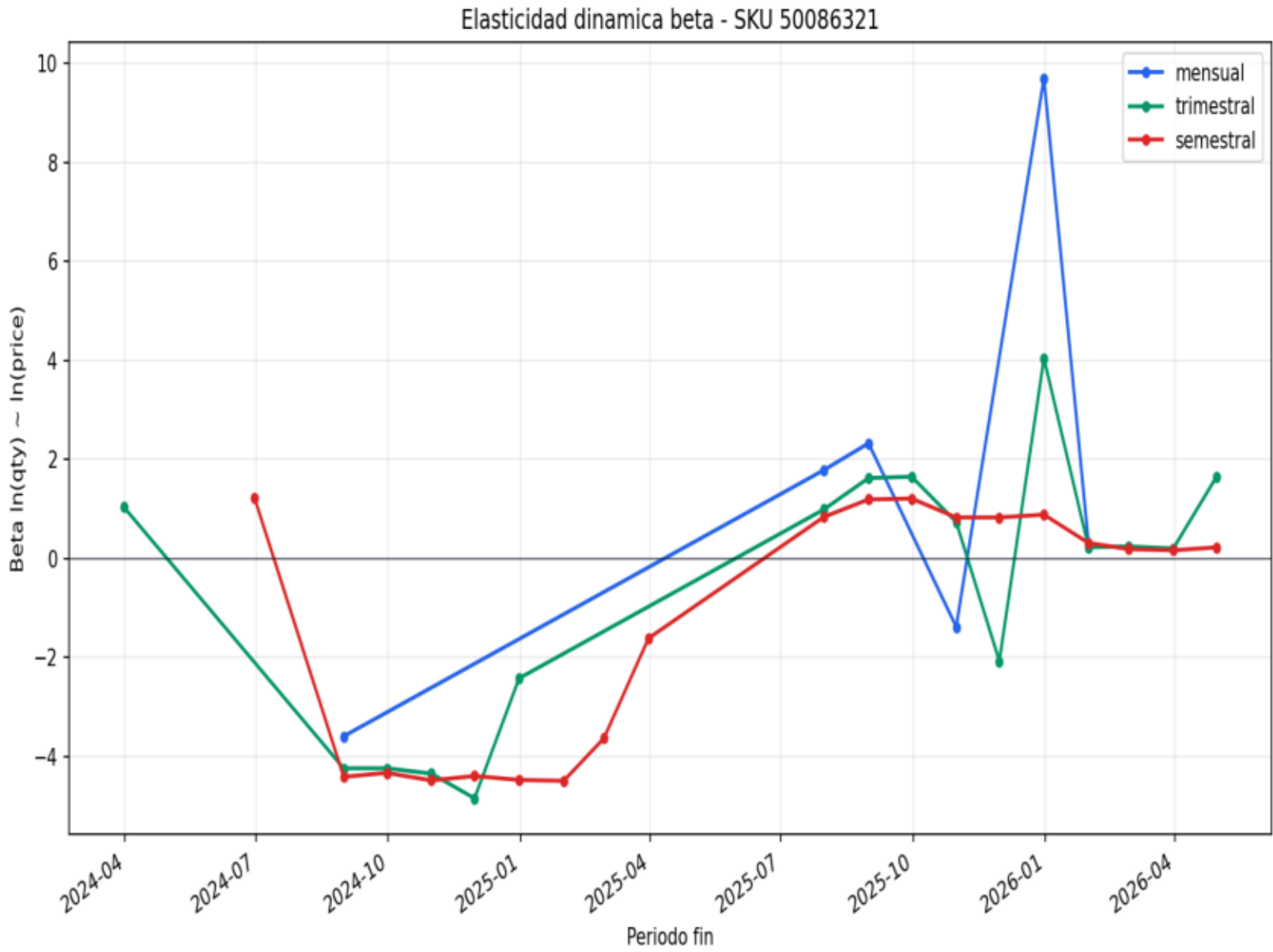
Beta mensual, trimestral y semestral - SKU 50077229



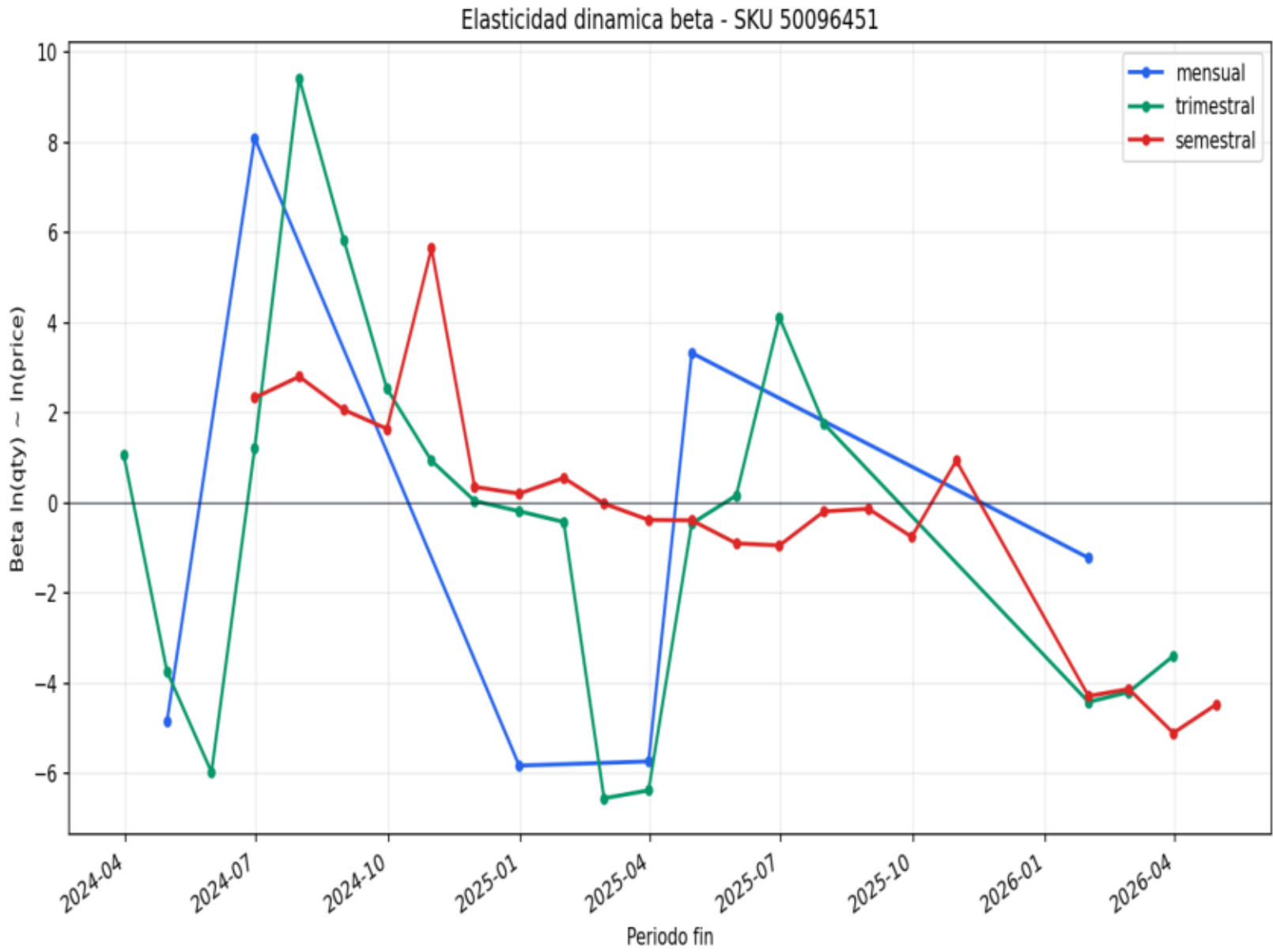
Beta mensual, trimestral y semestral - SKU 50084493



Beta mensual, trimestral y semestral - SKU 50086321



Beta mensual, trimestral y semestral - SKU 50096451



Beta mensual, trimestral y semestral - SKU 50106204

